The Binomial Theorem

From Factorials to Multinomial Expansion

A Mathematical Monograph

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February 25, 2026

Note For the document

This work is a personal study of the Binomial Theorem and its related topics, beginning from factorials and basic combinatorics and extending to the multinomial theorem and various applications.

These notes were written primarily for my own understanding and revision. They reflect my learning process, and therefore may contain errors, imprecisions, or omissions. I do not claim completeness or perfection.

If you happen to find any mistake, I would greatly appreciate your feedback. Please feel free to write to me at: `intmax2147@gmail.com`

Mathematics is learned through refinement, and I hope this document continues to improve over time.

Introduction to factorial

The factorial of natural number r is denoted by $r!$

Definition

Factorial of natural number r is the product of all natural number from 1 to r.

(a) $5! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$

(b) $3! = 1 \cdot 2 \cdot 3 = 6$

Observation

1. Factorial of fraction or negative numbers are not defined.
   (a) $(\frac{3}{4})! = undefined$
   (b) $(-12)! = undefined$
2. Factorial of every natural number is *even* except for 1.

Common Factorial Values

n	$n!$
1	1
2	2
3	6
4	24
5	120
6	720

Table 1: Common factorial values

Some Facts

1. Factorial of number greater than or equal to 5 will always have 0 in their unit place.
2. If a number is perfect square then only 6 of these are possible in unit place {0,1,4,5,6,9}
3. Sum of factorials after 4 will have unit place digit as 3.

Question: What is the minimum value of r if $r!$ is divisible by 13 where $r \in N$

Solution: $r! = 1 \cdot 2 \cdot 3 \cdots r$
In order to a number to be divisible by 13 it should have 13 in is factors, the only point the 13 is included in factorial of number when $r = 13$. So, $r_{min} = 13$.

Question: Is $13!$ divisible by 26

Solution: Both 13 and 2 comes in 13!. Also the factor of 26 is 2 and 13 hence, 13 is completely divisible by 26.

Question: Find the minimum value of $r(r \in N)$ if $r!$ is divisible by 16.

solution:
$1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \cdots$
Till here we have four 2's. From 2(1), 4(2),6(1), total four 2's we have.
And $16 = 2 \times 2 \times 2 \times 2$ Hence $r_{min} = 6$ which is completely divisible by 16.

Question: Solve for x, y where $x, y \in N$
$1! + 2! + 3! + \cdots + x! = y^2$

Solution: $r!$ will have unit digit as 0 for $r \geq 5$. Also the sum of factorial natural number for $r \geq 4$ will have unit digit 3. We know any number which is perfect square will not have unit place digit 3. Hence the equation will only satisfies for $x = 1$ and $x = 3$.

Question: Find the $x(\in N)$
such that it satisfy. $1! + 2! + 3! + \cdots + x! = 9 \cdot y$

Solution: The equation to satisfy the LHS should be multiple of 9.
Now the sum of factorial till 5 is divisible by 9.
$1! + 2! + 3! + 4! + 5! = 153$
The every factorial of number ≥ 6 is always be divisible by 9. As 3 comes two times till 6!. If a number is divisible by 9 and we add a number which is also divisible by 9 then the resulting number will always be divisible by 9.

Question: Evaluate the statement given below

Let

$$x = \sum_{r=1}^{2n} r!$$

$$y = \sum_{r=1}^{n} (2r)!$$

Statements

(a) x is divisible by y

(b) y is divisible by x

Solution:
$x = 1! + 2! + 3! + \cdots + (2n)!$
$y = 2! + 4! + 6! + \cdots + (2n)!$
Number of term in $x = 2n$ and
Number of term in $y = n$
Since number of term in y is less so y is never be divisible by x, that makes statement 1 false.
Now the x is odd, as after 2 all are even and when added 1 it become odd. And y is even because addition of factorial after 2 is even.
Odd can not divide even.

Exponent

Find the maximum value of $r(\in N)$ that $\frac{10!}{2^r}$ is an integer.

Solution:
$10! = 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9 \times 10$
Number of two's: $(2,1),(4,2),(6,1),(8,3),(10,1)$ So total 8 two's are there we can divide the 10!. Hence $r_{max} = 8$

General Formula

Exponent of prime p in $k!$ is given as. $\left[\frac{k}{p}\right] + \left[\frac{k}{p^2}\right] + \left[\frac{k}{p^3}\right] + \cdots$
Where [.] denotes G.I.F

Question: Exponent of 2 in 10!

Solution:
$\left[\frac{10}{2}\right] + \left[\frac{10}{4}\right] + \left[\frac{10}{16}\right] + \cdots$

$5 + 2 + 1 = 8$

Explanation

Let say we are find the maximum power of 3 which can divide the 100!.
$1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9 \times \cdots \times 27 \times \cdots \times 81 \times \cdots \times 99 \times 100$
When we are doing $\left[\frac{100}{3}\right]$ we get 33 which convey a meaning that there are 33 number which have 3 multiple. This is not sufficient, because 81 is although divisible by 3 but it counted just one power , we will use 9 and then sub sequence power of 3 to calculate the maximum power which can divide the given number. Just ask one simple question how many power does a number contribute in 3

Question: Find r_{max} if $120!$ is divisible by 6^r

Solution:
$6^r = 2^r \cdot 3^3$
We will find the 120! divisibility by 2^r and 3^r. The common power or the lower one will do the work.

Question: Find r_{max} for 12^r divides 120!

Answer is 58

Question: Find the number of zeros at the end of $150!$

Solution:

Since $10 = 2 \times 5$, we find the powers of 2 and 5 in 150!.

Power of 2:

$$\left[\frac{150}{2}\right] + \left[\frac{150}{4}\right] + \left[\frac{150}{8}\right] + \left[\frac{150}{16}\right] + \left[\frac{150}{32}\right] + \left[\frac{150}{64}\right] + \left[\frac{150}{128}\right] = 75 + 37 + 18 + 9 + 4 + 2 + 1$$
$$= 146.$$

Power of 5:

$$\left[\frac{150}{5}\right] + \left[\frac{150}{25}\right] + \left[\frac{150}{125}\right] = 30 + 6 + 1$$
$$= 37.$$

To form a factor of 10, we need one 2 and one 5. Hence, the number of trailing zeros is

$$\min(146, 37) = 37.$$

$$\boxed{37}$$

Question: Let $57 \cdot 58 \cdot 59 \cdots 200 = 5^r \cdot I$ Where $r, T \in N$, the find the r_{max}.

Solution:

$$57 \cdot 58 \cdot 59 \cdots 200 = \frac{1 \cdot 2 \cdot 3 \cdots 200}{1 \cdot 2 \cdot 3 \cdots 56} = \frac{120!}{56!}$$

Here we have to find the power of 5. The $r_{max} = 37$

1 Introduction to Binomial Coefficient

Binomial Coefficient is denoted by:

$$\boxed{{}^{n}C_r = \frac{n!}{r!(n-r)!}}$$

Example:

$${}^{10}C_3 = \frac{10!}{3!(10-3)!} = \frac{10!}{3! \times 7!} = 120$$

Observation:
Since factorial can never be negative,so

$$n - r \geq 0 \implies n \geq r$$

1.1 Meaning of Binomial Coefficient

The binomial coefficient represent number of ways in which r different objects can be selected from n different object. Here the position of object does not matter.

Example:

$${}^{4}C_2 = \frac{4!}{2! \times 2!} = 6$$

So there are 6 ways of choosing 2 different items from 4 distinct items. Lets verify.
Lets have $\{A, B, C, D\}$
Different combination of selection: $(AB), (BC), (CD), (DA), (AC), (BD)$
Also, $(AB) = (BA)$

1.2 More Interpretation:

We can also conclude the number of ways of selection is equal to number of ways of rejection. Suppose we have to select r object from n, either we can count the number of selecting r objects from n or we can select the $n - r$ object from n object.

1.3 Example

Let's take an example of selecting 3 different objects from 7 different objects. ${}^{7}C_3$
or we can say select 4 different object from 7 different object, that will be ${}^{7}C_4$

1.4 Formula

With above discussion we can conclude

$$\boxed{{}^{n}C_r = {}^{n}C_{n-r}}$$

1.5 Observation:

$$^{n}C_r = \frac{1 \cdot 2 \cdot 3 \cdots (n-3) \cdot (n-2) \cdot (n-1) \cdot n}{r!(n-r)!}$$

We can cancel terms from numerator till $n - r$, so

$$^{n}C_r = \frac{(n-r+1) \cdot (n-r+2) \cdots (n-3) \cdot (n-2) \cdot (n-1) \cdot n}{r!}$$

To count the number of terms in a range of value $lastTerm - firstTerm + 1$
So in the range,

$$\begin{aligned}(n-r+1) \cdot (n-r+2) \cdots (n-1) \cdot n &= \text{Last term} - \text{First term} + 1 \\ &= n - (n-r+1) + 1 \\ &= n - n + r - 1 + 1 \\ &= r.\end{aligned}$$

What we can observe that we have to expand the numerator r times and divide by $r!$

1.6 Observation

$$^{n}C_r = \frac{n!}{r!(n-r)!}$$

$$^{n}C_r = \frac{n \cdot (n-1)!}{r \cdot (r-1)!(n-r)!}$$

$$\boxed{^{n}C_r = \frac{n}{r} \times {}^{n-1}C_{r-1}}$$

1.7 Formula

- $\boxed{^{n}C_0 = {}^{n}C_n = 1}$
- $\boxed{^{n}C_1 = {}^{n}C_{n-1} = n}$

2 Introduction to Binomial Theorem

The rule with which rational exponent of sum of two variable can be expressed is called binomial Theorem.Expression containing two terms is called binomial.
Let $n \in N$, For any integer $n \in \mathbb{N}$ and any real numbers $x, y \in \mathbb{R}$,

$$(x+y)^n = \sum_{k=0}^{n} {}^nC_k x^{n-k} y^k$$

$$(x+y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \cdots + {}^nC_n y^n$$

2.1 Expansion Involving Negative Term

$$(x-2y)^2 = {}^nC_0 x^n + {}^nC_1 x^{n-1}(-2y) + {}^nC_2 x^{n-2}(-2y)^2 + \cdots + {}^nC_n(-2y)^n$$

2.2 Notes

- At any point of expansion the sum of exponent of both variable will always be n.
- Number of term in the expansion of $(x+y)^n$ will be $n+1$
- Coefficient equidistant from beginning and end will be equal.
- The kth term from the end is the $(n-k+2)^{\text{th}}$ term from the beginning.
- The kth term from the end in $(x+y)^n$ is equal to the kth term from the beginning in $(x+y)^n$.

2.3 Difference between Coefficient and Binomial Coefficient

Example 1: $(2x+3y)^7$

For $(a+b)^n$, the general term is

$$T_{r+1} = {}^nC_r a^{n-r} b^r.$$

Here, $a = 2x$, $b = 3y$, and $n = 7$.

General Term:

$$T_{r+1} = {}^7C_r (2x)^{7-r} (3y)^r$$

$$= {}^7C_r 2^{7-r} x^{7-r} 3^r y^r$$

$$= {}^7C_r 2^{7-r} 3^r \, x^{7-r} y^r.$$

Explanation:

$${}^7C_r$$

is called the **binomial coefficient**.

$${}^7C_r 2^{7-r} 3^r$$

is the **coefficient** of the term

$$x^{7-r} y^r.$$

2.4 Equidistant Coefficient

$$(x+y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \cdots + {}^nC_n y^n$$
$$(y+x)^n = {}^nC_0 y^n + {}^nC_1 y^{n-1} x + {}^nC_2 y^{n-2} x^2 + \cdots + {}^nC_n x^n$$

2.5 Some Common Expansion

1. $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$
2. $(1-x)^n = {}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 - {}^nC_3 x^3 + \cdots + (-1)^n [{}^nC_n x^n]$
3. $(x+a)^n + (x-a)^n = 2\,[{}^nC_0 x^n + {}^nC_2 x^{n-2} a^2 + {}^nC_4 x^{n-4} a^4 + \cdots]$
   (Only even powers of a appear.)
4. $(x+a)^n - (x-a)^n = 2\,[{}^nC_1 x^{n-1} a + {}^nC_3 x^{n-3} a^3 + {}^nC_5 x^{n-5} a^5 + \cdots]$
   (Only odd powers of a appear.)

2.6 General Term

We have seen the expansion of $(x+y)^n$ and $(y+x)^n$ are same, but when asked about the 20th term in the expansion of $(x+y)^n$, then this will not be same as the 20th term in the expansion of $(y+x)^n$
The T_{r+1} term in the expansion is given by

$$\boxed{T_{r+1} = {}^nC_r x^{n-r} y^r,}$$

where $r = 0, 1, 2, \ldots, n$.

Middle Term

In order to find middle term we need to calculate three values:

1. $\frac{TotalTerm}{2}$
2. $\frac{TotalTerm+1}{2}$
3. $\frac{TotalTerm+2}{2}$

Expression Multiplication

When two or more bracket in multiplication then:

1. At a time only one term in one bracket will take part in multiplication.
2. The Coefficient of final term will made up from the multiplication of individual Coefficient.

See the given expression below

$$(2x-3y)(x+4y)(x^2+1)(y-3)$$

$$(-3y)(4y)(x^2)(-3) = 36x^2y^2$$

Maximum Value of Binomial Coefficient

Suppose given expression is $(a+b)^n$. In the expansion of the expression

1. If $n \in even$, then $\frac{n}{2}$ will give maximum value in ${}^nC_r$.
2. If $n \in odd$, then $\frac{n+1}{2}$ and $\frac{n-1}{2}$ will give maximum value in ${}^nC_r$.

Excercise

Find the number of term in the expansion of following expression

1. $(x+y)^n$
2. $(x+y)^2$
3. $(x+2)^3$
4. $(x+y)^4$
5. $(x-y)^n$
6. $(x-y)^3$
7. $(2x+3y)^3$
8. $(x+y)^{10}$
9. $(x+y)^{15}$
10. $(2x-3y)^{10}$

Question

Simplify : ${}^{19}C_1 \cdot 3^5 \cdot 7^{35} + {}^{19}C_2 \cdot 3^6 \cdot 7^{33} + {}^{19}C_3 \cdot 3^7 \cdot 7^{31} + {}^{19}C_4 \cdot 3^8 \cdot 7^{28} \cdots$
Solution Lets talk about Arithmetic Progress (AP) first.
Let say a AP is given $3, 7, 11, 15, \cdots$.
How to write the general term of this AP
We observe the common difference is 4.So in general term $4r$ is confirm, now we have to decide from where we have to start the value of r, either start from 0 or start from 1. If we start from 0 the general term will be $4r+3$. If we want to start r from 1 then general term will be $4r-1$.

One more example of AP General Term
Let say the AP is: $120, 118, 116, 114 \cdots$
The common difference is -2, so in general term $-2r$ is to be confirmed. Now we have to decide wether we want r to start from 0 or 1. If we start r from 0 then general term will be, $-2r+120$. If we start r from 1 then general term will be $-2r+122$
This is done, let come to our question,
${}^{19}C_1 \cdot 3^5 \cdot 7^{35} + {}^{19}C_2 \cdot 3^6 \cdot 7^{33} + {}^{19}C_3 \cdot 3^7 \cdot 7^{31} + {}^{19}C_4 \cdot 3^8 \cdot 7^{28} \cdots$
The power of 3 follows a AP with common difference 1 and the power of 7 also follows AP with common difference -2, so general term for both will be, $(r+4)$ because r starts from 1, on similar line we have $(-2r+37)$ so the general term of the series given is
$T_r = {}^{19}C_r 3^{(r+4)} 7^{(-2r+37)}$

Question

Find the 6th term in the expansion of $\left(2x^2 - \frac{1}{3x^2}\right)^{10}$.
Solution

$$T_6 = {}^{10}C_5 \times (2x^2)^5 \times \left(\frac{-1}{3x^2}\right)^5$$

$$= {}^{10}C_5 \times 2^5 \times x^{10} \left(\frac{-1}{3}\right)^5 (\frac{1}{x^{10}})$$

$$= {}^{10}C_5 \times 2^5 \times \left(\frac{-1}{3}\right)^5$$

Question

The value of $(\sqrt{5}+1)^5 - (\sqrt{5}-1)^5$
Solution:

$${}^5C_1(\sqrt{5})^4 + {}^5C_3(\sqrt{5})^2 + {}^5C_5(\sqrt{5})^0 = 352$$

Question

In the expansion of $\left(x - \frac{1}{x}\right)^6$
Solution: For $r = 3$, the constant term will be$T_4 = -20$

Question

Find the constant term in $\left(3x^2 - \frac{5}{x}\right)^{20}$
Solution: For $r = 15$, the corresponding term will be constant.

3 Finding Coefficient of term

Question: Find the coefficient of following in the expression

1. x^2y^8 in $(x+y)^{10}$
2. x^2y^8 and x^3y^{12} in $(x+y)^{15}$
3. x^3y^7 in $(2x-3y)^{10}$
4. x^5y^10 in $(2x+9y)^{15}$
5. x^5, x^{-6}, x^{-5} in $\left(x+\frac{1}{x^3}\right)^{10}$

Question

Find the coefficient of x^{13} in $\left(2x+\frac{5}{x^2}\right)^{10}$
Solution: The coefficient is zero as no such term exist in the expansion. $r=\frac{7}{3}$, which is not valid.

Question

Find the coefficient of x^{11} in $\left(2x+\frac{5}{x^2}\right)^{20}$ Solution: $r=3$ will give the coefficient of required term.

Question

The value of

$$\frac{1}{81^n}-\frac{10}{81^n}{}^{2n}C_1+\frac{10^2}{81^n}{}^{2n}C_2-\frac{10^3}{81^n}{}^{2n}C_3+\cdots+10^{2n}$$

Solution

$$\begin{aligned}
&= \frac{1}{81^n}\left[1-10\cdot{}^{2n}C_1+10^2\cdot{}^{2n}C_2-10^3\cdot{}^{2n}C_3+\cdots+10^{2n}\right]\\
&= \frac{1}{81^n}\left[{}^{2n}C_0-10\cdot{}^{2n}C_1+10^2\cdot{}^{2n}C_2-10^3\cdot{}^{2n}C_3+\cdots+10^{2n}\cdot{}^{2n}C_{2n}\right]\\
&= \frac{1}{81^n}\left[\sum_{r=0}^{2n}{}^{2n}C_r10^r\right]\\
&= \frac{1}{81^n}(1-10)^{2n}\\
&= \frac{1}{9^{2n}}\times 9^{2n}\\
&= 1
\end{aligned}$$

Question

$(x+\sqrt{x^2-1})^6+(x-\sqrt{x^2-1})^6$ **Solution**
$2\cdot(32x^6-48x^4+18x^2-1)$

Question

The r_{th} term and $(r+4)^{th}$ are equal in the expansion of $(1+x)^{20}$. Then the value of r will be Solution

$$^{20}C_{r-1} = {}^{20}C_{r+3} =$$

$r = 9$

Question

Find the coefficient of term independent of x in the expansion of
$(1+x+2x^3)(\frac{3}{2}x^2 - \frac{1}{3x})^9$
Solution For $r = 6$ and $r = 7$ the term will be independent of x

Question

Find the term independent of$x$ in $(1+x)^m(1+\frac{1}{x})^n$ Solution

Question

In the expansion of $(x+a)^n$ if the sum of odd term be P and the sum of even term be Q then prove:

I. $P^2 - Q^2 = (x^2 - a^2)$

II. $4PQ = (x+a)^{2n} - (x-a)^{2n}$

Solution

$$
\begin{aligned}
(x+a)^n &= {}^nC_0x^n + {}^nC_1 \cdot x^{n-1}a^1 + {}^nC_2 \cdot x^{n-2}a^2 + \cdots \\
(x+a)^n &= T_1 + T_2 + T_3 + T_4 + T_5 + T_6 + T_7 \cdots \\
&= (T_1 + T_3 + T_5 + T_7 + T_9 \cdots) + (T_2 + T_4 + T_6 + T_8 + T_{10} \cdots) \\
&= P + Q
\end{aligned}
$$

$$
\begin{aligned}
(x-a)^n &= {}^nC_0x^n - {}^nC_1 \cdot x^{n-1}a^1{}^nC_2 \cdot x^{n-2}a^2 - \cdots \\
(x-a)^n &= T_1 - T_2 + T_3 - T_4 + T_5 - T_6 + T_7 \cdots \\
&= (T_1 + T_3 + T_5 + T_7 + T_9 \cdots) - (T_2 + T_4 + T_6 + T_8 + T_{10} \cdots) \\
&= P - Q \\
(x+a)^n &= P + Q \\
(x-a)^n &= P - Q
\end{aligned}
$$

$$
\begin{aligned}
(x+a)^n \cdot x-a)^n &= P^2 - Q^2 \\
[(x+a) \cdot x-a)]^n &= P^2 - Q^2 \\
(x^2 - a^2) &= P^2 - Q^2 \\
(x+a)^{2n} &= P^2 + Q^2 + 2PQ \\
(x-a)^{2n} &= P^2 + Q^2 - 2PQ \\
(x+a)^{2n} - (x-a)^{2n} &= 4PQ
\end{aligned}
$$

Question

Find the number of irrational terms in the expansion of $(4^{\frac{1}{5}} + 7^{\frac{1}{10}})^{45}$
Solution There are 41 irrational term.

Question

Find the number of rational term in the expansion of $(\sqrt[3]{2} + 7 \cdot \sqrt[5]{8})$
Solution: The number of rational term are 5, 20, 35

Question

if 21st **and** 22nd **terms in the expansion of** $(1-x)^{44}$ **equal then find** x
Solution: $x = \frac{-7}{8}$

Question

Find the value of n **if the ratio of** 7th **term from beginning and end in the expansion of** $\left(2^{\frac{1}{3}} + 3^{\frac{-1}{3}}\right)^n = \frac{1}{6}$
Solution : $n = 11$

Question

Find the coefficient of x^8 **in the expansion of** $(2 + 3x + 4x^3)$

Question

Find the Middle Term of the following:

I. $(2x - 3y)^{28}$

II. $\left(3x^2 - \frac{5}{x^3}\right)^{31}$

4 Greatest Binomial Coefficient

$$\begin{aligned}(1+x)^n &= {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_nx^n \\ &= {}^nC_0 + {}^nC_1 + {}^nC_2 \cdots + {}^nC_n\end{aligned}$$

For greatest binomial Coefficient

$$\begin{aligned}{}^nC_r &\geq {}^nC_{r-1} \\ \frac{n!}{r! \cdot (n-r)!} &\geq \frac{n!}{(r-1)! \cdot (n-r-1)!} \\ \frac{1}{r(n-r)} &\geq 1\end{aligned}$$

$$\boxed{r(n-r) \leq 1}$$

Notes

1. The middle term will be the greatest binomial coefficient.
2. If n is even it implies the number of binomial coefficient will be odd and there will be only one greatest binomial coefficient.
3. If n is odd then there will be even number of binomial coefficient hence there will be two greatest binomial coefficient and they both are equal.

Question

Find the value of r for which the following binomial Coefficient are maximum:

I. ${}^{50}C_r$

II. ${}^{57}C_r$

Question

Find r for which the binomial coefficient will be

1. Maximum
2. Minimum

Solution

$${}^rC_5 = \frac{r \cdot (r-1) \cdot (r-2) \cdot (r-3) \cdot (r-4) \cdot}{5!}$$

The fraction does not depend on the denominator, it depends on the value of r, hence as r increases the value will increase. The minimum value of r will be 5.

5 Numerically Greatest Term

5.1 Numerically Greatest Term

$$(x+y)^n = {}^nC_0x^n + {}^nC_1x^{n-1}y + {}^nC_2x^{n-2}y^2 + \cdots + {}^nC_ny^n$$

Numerically means without sign.
For example: $-6 > 4$

Method 1

Assume T_{r+1} is numerically greatest term, then

$$|T_r| \leq |T_{r+1}| \geq |T_{r+2}|$$

Method 2

$$\left(\frac{n+1}{1+\left|\frac{x}{y}\right|}\right) - 1 \leq r \leq \left(\frac{n+1}{1+\left|\frac{x}{y}\right|}\right)$$

Method 3

$$\begin{aligned}(x+y)^n &= \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \cdots + \binom{n}{n}y^n \\ &= T_1 + T_2 + T_3 + \cdots + T_{n+1}\end{aligned}$$

$$|T_{r+1}| \geq |T_r|$$

$$\left|\binom{n}{r}x^{n-r}y^r\right| \geq \left|\binom{n}{r-1}x^{n-r+1}y^{r-1}\right|$$

$$\binom{n}{r} = \frac{n-r+1}{r}\binom{n}{r-1}$$

$$\frac{n-r+1}{r}\,|x^{n-r}y^r| \geq |x^{n-r+1}y^{r-1}|$$

$$\frac{1}{r}\,|x^{n-r}y^r| \geq \frac{1}{n-r+1}\,|x^{n-r-1}y^{r-1}|$$

$$\boxed{r \leq \frac{(n+1)|xy|}{1-|xy|}}$$

5.2 For the expansion of $(1+x)^n$

$$\boxed{r \leq \frac{(n+1)|x|}{1+|x|}}$$

Question

Find the magnitude with greatest term in the expansion of of $(1+2x)^{25}$**, when** $x = \frac{-1}{5}$
Solution: $r \leq 7.4$ hence T_8 will be numerically greatest term.

Question

Find the greatest term in the expansion of $(4_5x)^{20}$ **where** $x = \frac{1}{2}$
Solution Upon solving the inequality $r \leq 15$, hence, T_{15} and T_{16} will be greatest term.

6 Properties of Binomial Coefficient

Property 1

$$\boxed{{}^nC_r = {}^nC_{r-1}}$$

Property 2

$$\boxed{{}^nC_r = Integer}$$

Property 3

If,

$$\boxed{{}^nC_x = {}^nC_y}$$

then, either

$$x = y$$
$$or$$
$$x + y = n$$

Property 4

$$\boxed{{}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}}$$
$$\boxed{{}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r}$$

Property 5

$$\boxed{{}^nC_r = \frac{n(n-1)(n-2)(n-3)\cdots(n-r+1)}{r!}}$$

Note: This implies that product of r natural number will always be divisible by $r!$

Property 6

$$\boxed{\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}}$$

Property 7

$$\boxed{r \cdot {}^nC_r = n \cdot {}^{n-1}C_{r-1}}$$

Property 8

when r in denominator

$$\boxed{\frac{{}^nC_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}}$$

Property 9

$$\boxed{{}^{n}C_{r} \cdot {}^{r}C_{k} = {}^{n}C_{k} \cdot {}^{n-k}C_{r-k}}$$

Property 10

Row Sum

Sum of all Binomial Coefficient = 2^n

$$ {}^{n}C_{0} + {}^{n}C_{1} + {}^{n}C_{2} + {}^{n}C_{3} + {}^{n}C_{4} + {}^{n}C_{5} + \cdots + {}^{n}C_{n} = 2^n $$

$$ \sum_{n}^{r=0} {}^{n}C_{r} = 2^n $$

Generalized Row Sum

$$ \sum_{n_1+n_2+n_3+\cdots n_r=n} \frac{n!}{(n_1)!(n_2)!(n_3)!\cdots n_r!} = r^n $$

Property 11

$$ \sum_{r=0}^{n} (-1)^n \times {}^{n}C_{r} = 0 $$

Property 12

Row Square Sum

$$ \sum_{r=0}^{n} \left({}^{n}C_{r}\right)^2 = {}^{2n}C_{r} $$

Property 13

Column Sum

$$ \sum_{k=0}^{n} {}^{k}C_{r} = {}^{n+1}C_{r+1} $$

Property 14

Sum of all odd binomial coefficient = Sum of all even binomial coefficient = 2^{n-1}

$$ {}^{n}C_{1} + {}^{n}C_{3} + {}^{n}C_{5} + {}^{n}C_{7} + \cdots = 2^{n-1} $$

$$ {}^{n}C_{0} + {}^{n}C_{2} + {}^{n}C_{4} + {}^{n}C_{6} + \cdots = 2^{n-1} $$

Property 15

Vamdermonde's Identity

$$ \sum_{r=0}^{k} {}^{m}C_{r} \times {}^{n}C_{k-r} = {}^{m+n}C_{ks} $$

Property 16

Hockey Stick Property

$$^{k}C_k + {}^{k+1}C_k + {}^{k+2}C_k + \cdots + {}^{k+n}C_k = {}^{k+n+1}C_{k+1}$$

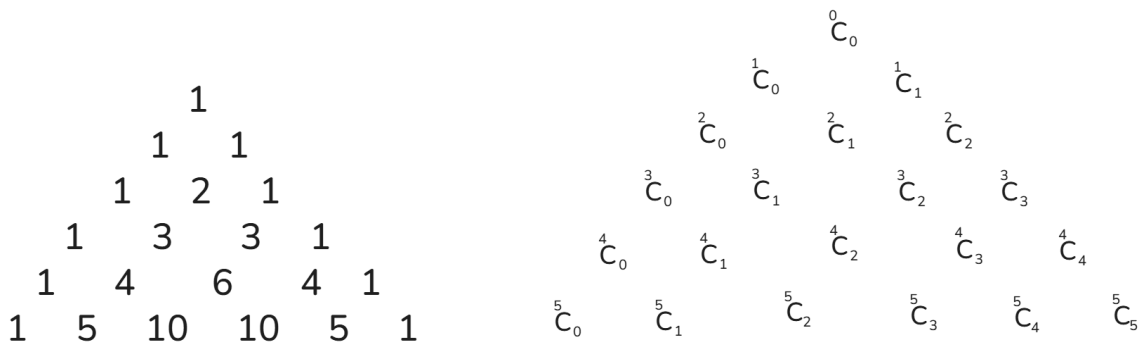

Figure 1: Pascal Triangle

7 Problem Based on Property

7.1 Question

Prove that $(n^2)!$ is divisible by $(n!)^n$
Solution:

$$(n!)^2 = 1 \cdot 2 \cdot 3 \cdots n^2$$
$$(n!)^2 = 1 \cdot 2 \cdot 3 \cdots n \cdot (n+1) \cdots 2n \cdots 3n \cdots (n-1)^2 \cdot n^2$$
$$= (1 \cdot 2 \cdot 3 \cdots n)(n+1 \cdots 2n)(2n+1 \cdots 3n) \cdots (\cdots n^2)$$

All the grouping are n natural number. Since all are consecutive n natural number in each group. So each group must be divisible by $n!$ using property 4

7.2 Question

Find the value of the expression :

$$\frac{{}^{20}C_1}{{}^{20}C_0} + 2 \cdot \frac{{}^{20}C_2}{{}^{20}C_1} + 3 \cdot \frac{{}^{20}C_3}{{}^{20}C_2} + \cdots + 20 \cdot \frac{20C_{20}}{{}^{20}C_{19}}$$

Solution
The Expression can be written in compact for as,

$$\sum_{r=1}^{20} r \cdot \frac{{}^{20}C_r}{{}^{20}C_{r-1}}$$
$$property5$$
$$\sum_{r=1}^{20} r \cdot \frac{n-r+1}{r}$$
$$\sum_{r=1}^{20} (n-r+1)$$
$$= 210$$

7.3 Question

Find the λ:

$$({}^nC_0 + {}^nC_1) + ({}^nC_1 + {}^nC_2) + ({}^nC_2 + {}^nC_3) + \cdot({}^nC_{n-1} + {}^nC_n) = \lambda \cdot {}^nC_1 \cdot {}^nC_2 \cdot {}^nC_2 \cdots {}^nC_n$$

Solution: $\lambda = \frac{(n+1)^2}{n!}$

7.4 Question

Find the value of :

$$^{19}C_0 + {}^{19}C_1 \cdot {}^{19}C_2 \cdots {}^{19}C_9$$

Solution: 2^{18}

7.5 Question

$$^{30}C_1 + {}^{30}C_1 \cdot {}^{30}C_2 \cdots {}^{30}C_{14} = ?$$

8 Series sum in binomial Theorem

8.1 Type 1 : When term containing r is present in numerator.

Method : Use $r \cdot {}^nC_r$ formula to get rid of term containing r and then use sum of binomial coefficient.

Question 1: Find the sum of the series

$$1 \cdot {}^nC_0 + 2 \cdot {}^nC_1 + 3 \cdot {}^nC_2 + 4 \cdot {}^nC_3 + \cdots$$

Solution: $2^{n-1} \cdot (n+1)$

Question 2: Find the sum of the series

$$5 \cdot {}^{17}C_1 + 8 \cdot {}^{17}C_2 + 11 \cdot {}^{17}C_3 + \cdots$$

Solution: $51 \cdot 2^{16} + 2 \cdot 2^{17} - 2$

Results

$$\boxed{\sum_{r=0}^{n} \cdot {}^nC_r = 2^n}$$

$$\boxed{\sum_{r=0}^{n} r \cdot {}^nC_r = 2^{n-1}}$$

$$\boxed{\sum_{r=0}^{n} r(r-1) \cdot {}^nC_r = n \cdot (n-1) \cdot 2^{n-1}}$$

$$\boxed{\sum_{r=0}^{n} r(r-1)(r-2) \cdot {}^nC_r = n \cdot (n-1) \cdot (n-2) \cdot 2^{n-1}}$$

$$\boxed{\sum_{r=0}^{n} r^2 \cdot {}^nC_r = n(n-1)2^{n-1} + n \cdot 2^{n-1}}$$

Question:

$$1 \cdot 3 \cdot {}^{20}C_1 + 3 \cdot 7 \cdot {}^{20}C_2 + 5 \cdot 11 \cdot {}^{20}C_3 \cdots$$

Solution: $8(20 \cdot \ 19 \cdot 2^{18}) + 5 \cdot 2^{22} + 2^{20} - 1$

8.2 Series sum of negative term

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$$
$$x = -1$$
$$0 = {}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 - \cdots + (-1)^n \cdot {}^nC_n x^n$$
$$\boxed{{}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 - \cdots + (-1)^n \cdot {}^nC_n x^n = 0}$$

Question:

$${}^{30}C_0 \cdot 5^{100} + {}^{30}C_1 \cdot 5^{98} \cdot 3^3 + {}^{30}C_2 \cdot 5^{96} \cdot 3^6 + \cdots$$

Solution: $2^{30} \cdot 5^{40}$

Question:

$$1 \cdot {}^{50}C_1 - 4 \cdot {}^{50}C_2 \cdot 5^2 + 7 \cdot {}^{50}C_3 \cdot 5^4 - 10 \cdot {}^{50}C_4 \cdot 5^6$$

8.3 Type 2: When term containing r in the denominator

1. use $\frac{{}^nC_r}{r+1}$
2. Use Integration

Question:

$$\frac{{}^{25}C_1}{2} + \frac{{}^{25}C_2}{3} + \frac{{}^{25}C_3}{4} \cdots + \frac{{}^{25}C_{25}}{26}$$

Solution: $\frac{2^{26}-27}{26}$

Question:

$$\frac{{}^{40}C_0}{2} + \frac{{}^{40}C_1}{3} + \frac{{}^{40}C_2}{4} \cdots + \frac{{}^{40}C_{39}}{41}$$

Solution: Find the compact form and then multiply and divide it by $(r+1)$

Question:

$$\frac{{}^{32}C_0}{1 \cdot 2} + \frac{{}^{32}C_2}{2 \cdot 3} + \frac{{}^{32}C_2}{3 \cdot 4} \cdots$$

Question

$$\frac{{}^{17}C_0}{1 \cdot 2 \cdot 3} \cdot 4^5 - \frac{{}^{17}C_1}{2 \cdot 3 \cdot 4} \cdot 4^7 + \frac{{}^{17}C_2}{3 \cdot 4 \cdot 5} \cdot 4^9 - \cdots$$

Question:

$$\frac{{}^nC_0}{n} - \frac{{}^nC_1}{n+1} + \frac{{}^nC_2}{n+2} - \frac{{}^nC_3}{n+3} + \cdots$$

Solution: we know

$$(1-x)^n = {}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 - {}^nC_3 x^3 + \cdots$$

Multiply the above equation by x^{n-1} and integrate from 0 to 1.

Question

$$\frac{{}^nC_0}{(n+2)(n+3)} - \frac{{}^nC_1}{(n+3)(n+4)} + \frac{{}^nC_2}{(n+4)(n+5)} - \cdots$$

8.4 Type 3: When series contain product of two binomial coefficient

If the sum of lower sub-script of two binomial coefficient in general term is constant or can be made constant then multiply corresponding binomial expansion and compare appropriate coefficient on both side.

Question Lecture Number: 68

$$^{n}C_0 \cdot {}^{n+2}C_3 + {}^{n}C_1 \cdot {}^{n+2}C_4 + {}^{n}C_2 \cdot {}^{n+2}C_5 + \cdots$$

Solution: $^{2n+2}C_{n-1}$

Question Lecture 69

$$^{n-3}C_4.^{2n+1}C_2 + {}^{n-3}C_5.^{2n+1}C_3 + {}^{n-3}C_6.^{2n+1}C_4 + \cdots$$

Question: Lecture 70

$$^{n}C_0 \cdot {}^{2n}C_0 + {}^{n}C_1 \cdot {}^{2n}C_1 + {}^{n}C_2 \cdot {}^{2n}C_2 + \cdots$$

Question: Lecture 70

$$^{n}C_n \cdot {}^{n}C_1 + {}^{n}C_{n-1} \cdot {}^{n}C_2 + {}^{n}C_{n-2} \cdot {}^{n}C_3 + \cdots$$

Question: Lecture 71

$$^{n}C_n \cdot {}^{n+1}C_1 + {}^{n}C_{n-1} \cdot {}^{n+1}C_2 + {}^{n}C_{n-2} \cdot {}^{n+1}C_3 + \cdots$$

Question: Lecture 72

$$1 \cdot {}^{n}C_0 \cdot {}^{n}C_2 + 2 \cdot {}^{n}C_1 \cdot {}^{n}C_3 + 3 \cdot {}^{n}C_2 \cdot {}^{n}C_4 + \cdots$$

Question

$$(1 + x + x^2)^n = a_0 + a_1 \cdot x^1 + a_2 \cdot x^2 + a_3 \cdot x^3 + \cdots a_{2n} \cdot x^{2n}$$

the fin the sum

$$a_0 \cdot {}^{n}C_0 - a_1 \cdot {}^{n}C_1 + a_2 \cdot {}^{n}C_2 - a_3 \cdot {}^{n}C_3 + \cdots$$

8.5 Type 4: When lower subscript of binomial coefficient is constant

Procedure: Replace the term with corresponding expansion then manipulate.

Example:
Coefficient of x^12 in $(x+1)^{25} = {}^{25}C_{12}$,
or we can also say
${}^{25}C_{12} =$ Coefficient of x^12 in $(x+1)^{25}$

Exmaple:
${}^{41}C_{18} + {}^{31}C_{18} =$ Coefficient of x^{18} in $(x+1)^{41}$ + Coefficient of x^{18} in $(x+1)^{31}$
Coefficient of x^{18} in $((1+x)^{41} + (1+x)^{31})$

Exmaple:

$$ {}^{30}C_7 + {}^{31}C_7 + {}^{41}C_7 $$

$=$ Coefficient of x^7 in $((1+x)^{30} + (1+x)^{31} + (1+x)^{41})$
$=$ Coefficient of x^7 in $((1+x)^{30} + (1+x)^{31} + (1+x)^{41} + (1+x)^6)$
Adding lower power will not affect the coefficient.

Question: Find

$$ {}^{k}C_k + {}^{k+1}C_k + {}^{k+2}C_k + \cdots {}^{k+n}C_k $$

Solution:
Coefficient of x^k in $((1+x)^k + (1+x)^{k+1} + (1+x)^{k+2} + \cdots + (1+x)^{k+n})$

Coefficient of x^k in $((1+x)^k + (1+x)^{k+1} + (1+x)^{k+2} + \cdots + (1+x)^{k+n} + (1+x)^{k-1} + (1+x)^{k-2} + (1+x)^{k-3} + = \cdots + (1+x) + 1)$

Coefficient of x^k in $(1+(1+x)+(1+x)^2+(1+x)^3+(1+x)^4+(1+x)^5+\cdots(1+x)^k+\cdots+(1+x)^n)$
The above expression become geometric progression we can find the sum using GP

Question: Lecture 75

$$ {}^{n}C_0 \cdot {}^{n}C_n - {}^{n}C_1 \cdot {}^{2n}C_n + {}^{n}C_2 \cdot {}^{3n}C_n - \cdots $$

9 Multinomial Theorem

9.1 Introduction

Coefficient of $x^{n_1} \cdot y^{n_2} \cdot z^{n_3}$ in the expansion of $(k_1 \cdot x + k_2 \cdot y + k_3 \cdot z)^n$

1. 0 if $n_1 + n_2 + n_3 \neq n$
2. $\left(\frac{n!}{n_1! \cdot n_2! \cdot n_3!}\right) \cdot k_1^{n_1} \cdot k_2^{n_2} \cdot k_3^{n_3}$
   if $n_1 + n_2 + n_3 = n$

9.2 Trinomial

$$\begin{aligned}
(x+y+z)^n &= (x+y+z)(x+y+z)(x+y+z)\cdots(x+y+z) \\
&= {}^{n}C_{n_1} x^{n_1} \times {}^{n-1}C_{n_2} y^{n_2} \times {}^{n-n_1-n_2}C_{n-n_1-n_2} z^{n-n_1-n_2} \\
&= \frac{n!}{(n-n_1)!n_1!} \cdot x^{n_1} \times \frac{(n-n_1)!}{(n-n_1-n_2)!(n-n_1)!} \cdot y^{n_2} \times z^{n_3} \\
&= x^{n_1} \cdot y^{n_2} \cdot z^{n_3} \times \frac{n!}{n_1!n_2!n_3!} \\
&= \frac{n!}{n_1!n_2!n_3!} \times x^{n_1} \cdot y^{n_2} \cdot z^{n_3}
\end{aligned}$$

Question: In the expansion of $(3x - 4y + 8z)^{25}$ coefficient of:

1. $x^2y^3z^7$
2. $x^{19}y^2z^4$
3. $z^{21}y^4$

Question: Find the coefficient of x^5 in the expansion of $(4 - 3x + 7x^2)$

Question: Expand $(x + y + z)^2$

Question: Find the number of term in the following expansion

1. $(x + y + z)^{10}$
2. $(x + y + z)^{15}$
3. $(x + y + z + w + a)^{20}$

Question Find the coefficient of :

1. $x^2y^3z^5$ in the expansion of $(x + y + z)^{10}$
2. $x^2y^3z^{10}$ in the expansion of $(x + y + z)^{15}$

Question Find the coefficient of following int the expansion of $(2x - 3y + 4z)^{10}$

1. $x^2y^3z^5$
2. $x^3y^4z^5$

9.3 Number of term

Numbe of term in the expansion of

$$(x_1 + x_2 + x_3 + x_4 + \cdots + x_k)^n$$

will be ${}^{x+k-1}C_{k-1}$. All variable should be different.

Question: Number of term in the expansion of $(x + y + z)^{20}$

9.4 Binomial Theorem for Rational and Negative Integer

$$(1 + n)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots + \infty$$

$n = RationalNumber/NegativeInetger$
$|n| < 1$

10 Application of Binomial Theorem

10.1 Last Digit Repetition Cycle

The unit digit of n^m repeats cyclically as m increases. The cycle depends only on the unit digit of n.

Base (Unit Digit)	Repeating Cycle	Cycle Length
1	1	1
2	2,4,8,6	4
3	3,9,7,1	4
4	4,6	2
5	5	1
6	6	1
7	7,9,3,1	4
8	8,4,2,6	4
9	9,1	2

Table 2: Unit Digit Cycles of Powers

Example: The last digit of following number

1. $2^{4m} \to 6$
2. $2^{4m+1} \to 2$
3. $2^{4m+2} \to 64$

Question: What will be last digit of expression

$$2222^{5555} + 7777^{3333}$$

Solution:
Since 2 and 7 have repeat cycle of 4 hence the power will be represented in the multiple of 4

$$\begin{aligned}
&= 2222^{5555} + 7777^{3333} \\
&= 2222^{4m+3} + 7777^{4m+1} \\
&= 8 + 7 \\
&= 15 \\
&= 5
\end{aligned}$$

10.2 Last two digit

To find last digit of n^m express base in term of $(10k \pm 1)$ then expand. It will work when unit digit of x is $[1, 3, 7, 9]$

Question: Find last two digit of 3^{73}.
Solution:

$$\begin{aligned}
&= 3 \cdot 3^{72} \\
&= 3 \cdot (3^2)^{36} \\
&= 3 \cdot (10-1)^{36} \\
&= 3 \cdot ({}^{36}C_0 \cdot 10^{36} + {}^{36}C_1 \cdot 10^{35}{}^{36}C_2 + \cdot 10^{34} \cdots + {}^{36}C_{34} \cdot 10^2 - {}^{36}C_{35} + {}^{36}C_{36})
\end{aligned}$$

When we have 10^2 then last digit will be 00 so we just have to add the rest numbers.

Question: Find the last two digit of 2^{51}
Solution: Break the number in $(30+2)$ and expand.

10.3 Divisibility Test

If a natural number n is divisible by m, the remainder can take value from $\{0, 1, 2, 3, \cdots, m-1\}$

$R\left(\frac{n}{m}\right) = k$, then $k \in \{0, 1, 2, 3, \cdots, m-1\}$

$R\left(\frac{m \cdot n}{m}\right) = R\left(\frac{n}{m}\right)$

Formula 1:

$$(x^n + y^n) = (x+y)\left[x^{n-1} - x^{n-2} \cdot y^1 + x^{n-3} \cdot y^2 - x^{n-4} \cdot y^3 + x^{n-5} \cdot y^4 \cdots + y^{n-1}\right]$$

This formula is applicable only when n is odd

Formula 1:

$$(x^n - y^n) = (x-y)\left[x^{n-1} - x^{n-2} \cdot y^1 + x^{n-3} \cdot y^2 + x^{n-4} \cdot y^3 + x^{n-5} \cdot y^4 \cdots + y^{n-1}\right]$$

Question: Check whether $17^{51} + 8^{51}$ is divisible by 5.

Question: Check whether $17^{30} + 8^{30}$ is divisible by 9.

Question: Check whether $20^{45} + 50^{91}$ is divisible by 7. Further check whether it is divisible by 14.

10.4 Remainder:

To find remainder when n^m is divided by p. List exponent of n and multiple of p and then identify two number which has difference of 1. Then to find required remainder express n^m in term of identified number.

Question: Find the remainder when 2^{157} when divide by 7.

Solution:
$p = 7$
$n^m = 2^{157}$, $n = 2, m = 157$

Power of 2	Multiple of 7
2	7
4	14
8	21
16	28
32	35
64	42

Table 3: Relation between powers of 2 and multiples of 7

Now you can see (8, 7) have the difference of 1.

$$\begin{aligned} &= 2^{157} \\ &= 2 \cdot (2^3)^5 2 \\ &= 2 \cdot (8)^5 2 \\ &= 2 \cdot (1+7)^5 2 \end{aligned}$$

Here we can expand $(1+7)^{52}$. The terms will have power 7. Which is absouletly didivible by 7. Some term will not have, add them and find the remainder.

Question: $R\left(\frac{(123)^{85}}{11}\right)$

11 Problems

Problem 1

For

$$2 \leq r \leq n, {}^{n}C_r + 2 \cdot {}^{n}C_{r-1} + {}^{n}C_{r-2} =$$

(A) ${}^{n+1}C_{r-1}$

(B) ${}^{n+1}C_{r+1}$

(C) ${}^{n+2}C_r$

(D) ${}^{n+2}C_r$

Problem 2

Find the coefficient of x^{49} in the polynomial

$$\left(x - \frac{C_1}{C_0}\right)\left(x - 2^2 \cdot \frac{C_2}{C_1}\right)\left(x - 3^2 \cdot \frac{C_3}{C_2} \cdots\right)\left(x - 50^2 \cdot \frac{C_{50}}{C_{49}}\right)$$

Where $C_r = {}^{50}C_r$

Problem 3

The sum

$$\sum_{i=0}^{m} {}^{10}C_i \cdot {}^{20}C_{m-i}$$

Where ${}^{p}C_q = 0$ if $p < q$
The sum is maximum when m is

(A) 5

(B) 10

(C) 15

(D) 20

Problem 4:

Prove that

$$2^k \cdot {}^{n}C_0 \cdot {}^{n}C_k - 2^{k-1} \cdot {}^{n}C_1 \cdot {}^{n-1}C_{k-1} + 2^{k-2} \cdot {}^{n}C_2 \cdot {}^{n-2}C_{k-2} \cdots (-1)^n \cdot {}^{n}C_k \cdot {}^{n-k}C_0 = {}^{n}C_k$$

Problem 5

For any positive integer m, n (with $n \geq m$),Prove that,

$$^{n}C_m + {}^{n-1}C_m + {}^{n-2}C_m + \cdots + {}^{n-m}C_m = {}^{n+1}C_{m+1}$$

Hence or otherwise prove that

$$^{n}C_m + 2 \cdot {}^{n-1}C_m + 3 \cdot {}^{n-2}C_m + \cdots + (n-m+1) \cdot {}^{n-m}C_m = {}^{n+1}C_{m+1}$$

12 Problems on the Binomial Theorem

Question 1
(JEE Mains 2024)

Let m and n be the coefficients of the 7th and 13th terms respectively in the expansion of

$$\left(\frac{1}{3}x^{\frac{1}{3}} + \frac{1}{2x^{\frac{2}{3}}}\right)^{18}.$$

Find $\left(\frac{n}{m}\right)^{\frac{1}{3}}$.

Question 2
(JEE Mains 2022)

If the sum of the coefficients of all positive even power of x in the binomial expansion of $(2x^3 + \frac{x}{3})^{10}$ is $5^{10} - \beta.3^9$, then β is equal to

Question 3
(JEE Adv 2013, JEE Mains 2020)

Coefficient of three consecutive term in the expansion of $(1+x)^{n+5}$ are in ratio $5:10:14$ then n is equal to?

Question 4
(JEE Mains 2024)

Let the coefficient of x^r in the expansion of

$$(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + (x+3)^{n-4}(x+2)^3 + ... + (x+2)^{n-1}$$

be a_r. If $\sum_{r=0}^{n} a_r = \beta^n - \gamma^n, \beta, \gamma \epsilon N$, then the value of $\beta^n + \gamma^n =$?

Question 5
(JEE Mains 2022)

If the coefficient of x^{10} in the binomial expansion of

$$(\frac{\sqrt{x}}{5^{\frac{1}{4}}} + \frac{\sqrt{5}}{x^{\frac{1}{3}}})$$

is $5^k l$, where $l, k\epsilon N$ and l is co-prime to 5, then k equal to =?

Question 6

The coefficient of x^{301} in

$$(1+x)^{500} + x(1+x)499 + x^2(1+x)^{498} + ... + x^{500}$$

is?

Question 7

The number of integral terms in the expansion of

$$(7^{(\frac{1}{2})} + 11^{(\frac{1}{6})})^{824}$$

Question 8
(JEE Mains 2019)

The total number of irrational terms in the binomial expansion of

$$(7^{\frac{1}{5}} - 3^{\frac{1}{10}})^{60}$$

Question 9
(JEE Mains 2022)

The term independent of x in the expression of

$$(1 - x^2 - 3x^3)(\frac{5}{2}x^3 - \frac{1}{5x^2})^{11}$$

, $x \neq 0$

Question 10
(JEE Mains 2023)

If the coefficient of x and x^2 in $(1+x)^p(1-x)^q$ are 4 and -5 respectively, then $2p+3q$ equal to?

Question 11
(JEE Mains 2023)

If the constant term in the expansion of given expression is p, then $108p$ equals to?

$$(1+2x-3x^2)(\frac{3}{2}x^2-\frac{1}{3x})^9$$

Question 12
(JEE Mains 2024)

In the expansion of

$$(1+x)(1-x^2)(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3})^5, x \neq 0,$$

then sum of the coefficient of x^3 and $x(-13)$ is equal to?

Question 13
(JEE Mains 2024)

If A denotes the sum of all the coefficient in the expansion of $(1=3x+10x^2)^n$ and B denotes the sum of all the coefficient in the expansion of $(1+x^2)$, then:

1. $B = A^3$
2. $3A = B$
3. $A = 3B$
4. $A = B^3$

Question 14
(JEE Mains 2019)

The coefficient of x^{18} in the product of

$$(1+x)(1-x)^{10}(1+x+x^2)^9$$

Question 15
(JEE Mains 2021)

Let n be the positive integer. Let

$$A = \sum_{k=0}^{n}(-1)^k[(\frac{1}{2})^k+(\frac{3}{4})^k+(\frac{7}{8})^k+(\frac{15}{16})^k+(\frac{31}{32})^k]$$

If $63A = 1-\frac{1}{2^{30}}$ then n is equal to?

Question 16
(JEE Mains 2022)

Let for the 9th term in the binomial expansion of $(3+6x)^n$, in the increasing power of $6x$, to be the greatest for $x = \frac{3}{2}$, the least value of n is n_o. If k is the ration of the coefficient of x^6 to the coefficient of x^3, then $k + n_o$ is equal to.

Question 17
(JEE Mains 2024)

$$^{n-1}C_r = (k^2 - 8)^n C_{r+1}$$

If and only if:

1. $2\sqrt{2} < k \leq 3$
2. $2\sqrt{3} < k \leq 3\sqrt{2}$
3. $2\sqrt{3} < k < 3\sqrt{3}$
4. $2\sqrt{2}k < k < 2\sqrt{3}$

Question 18
(JEE Mains 2020)

The value of $\sum_{r=0}^{20} {}^{50-r}C_{r+6}$ is equal to:

1. ${}^{50}C_6 - {}^{30}C_6$
2. ${}^{51}C_7 + {}^{30}C_7$
3. ${}^{51}C_7 - {}^{30}C_7$
4. ${}^{50}C_7 - {}^{30}C_7$

Question 19
(JEE Mains 2021)

Let ${}^nC_r$ denote the binomial coefficient of x^r in the expansion of $(1+x)^n$. If

$$\sum_{k=0}^{10} (2^k + 3k)^{10}C_k = \alpha.3^{10} + \beta.2^{10}, \alpha, \beta \in R$$

Then $\alpha + \beta =?$

Question 20
(JEE Mains 2023)

Suppose

$$\sum_{r=0}^{2023} r^2 . {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}.$$

Then the value of α is.

Question 21
(JEE Mains 2023)

If,

$$\frac{1}{n+1}{}^nC_n + \frac{1}{n}{}^nC_{n-1} + \cdots + {}^nC_0 = \frac{1023}{10}$$

. Then n equal to.

Question 22
(JEE Mains 2024)

Let

$$\alpha = \sum_{k=0}^{n} (\frac{({}^nC_k)^2}{k+1})$$

$$\beta = \sum_{k=0}^{n-1} (\frac{({}^nC_k)({}^nC_{k+1})}{k+2})$$

If $5\alpha = 6\beta$, then n equals.

Question 23
(JEE Mains 2019)

The value of r for which

$${}^{20}C_r \cdot {}^{20}C_0 + {}^{20}C_{r-1} \cdot {}^{20}C_1 + {}^{20}C_{r-2} \cdot {}^{20}C_2 + \cdots + {}^{20}C_0 \cdot {}^{20}C_r$$

is maximum

Question 24
(JEE Adv. 2018)

Let ,

$$X = ({}^{10}C_1)^2 + 2({}^{10}C_2)^2 + \cdots + 10({}^{10}C_{10})^2$$

Where $r \in \{1, 2, 3. \cdots, 10\}$ and ${}^{10}C_r$ denote binomial coefficient. Then value of $\frac{1}{1430}X$ is?

Question 25
(JEE Mains. 2020)

Let

$$(2X^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r.$$

Then $\frac{a_7}{a_{13}}$ is equal to?

Question 17

$$P = \sum_{j=0}^{n} \sum_{i=0}^{n} ({}^nC_i + {}^nC_j)$$

$$Q = \sum_{j=0}^{n} \sum_{i=0}^{n} ({}^nC_i + {}^nC_j)$$

Where $i = j$

$$R = \sum_{0 \le i < j \le n}^{n} \sum^{n} ({}^nC_i + {}^nC_j)$$

$$S = \sum_{0 \le i \le j \le n}^{n} \sum^{n} ({}^nC_i + {}^nC_j)$$

Question 26
(JEE Mains. 2022)

$$\sum_{i,j=0}^{n} {}^nC_i \cdot {}^nC_j$$

is equal to

1. $2^{2n} - {}^{2n}C_n$
2. $2^{2n-1} - {}^{2n-1}C_{n-1}$
3. $2^{2n} - \frac{1}{2} \cdot {}^{2n-1}C_{n-1}$
4. $2^{2n} + {}^{2n-1}C_{n-1}$

Question 27

In the expansion if $(1 + x + x^2)^5$ find:

1. Coefficient of x^4
2. Number of terms

Question 28
(JEE Mains. 2021)

If the coefficient of a^7b^8 in the expansion of $(a+2b+4ab)^{10}$ is $K.2^{16}$, then K is equal to?

Question 29
(JEE Mains. 2023)

Let,

$$(a+b+cx^2) = \sum_{i=0}^{20} p_i x^i$$

$$a, b, c \in N$$

. If $p_1 = 20$ and $p_2 = 210$ then $2(a+b+c)$ is equal to

Question 30
(JEE Mains. 2023)

The coefficient of x^7 in $(1-x+2x^3)^{10}$ is

Question 31

Find the remainder for the following.

1. $7^{2022} + 3^{2022}$ divided by 5.
2. $\{\frac{2^{403}}{15}\}$, Where $\{.\} \equiv$
3. $\{\frac{2^{200}}{8}\}$, Where $\{.\} \equiv$
4. $(11)^{1011} + (1011)^{11}$ is divided by 9

Question 32
(JEE Mains. 2024)

Remainder when $64^{32^{32}}$ is divided by 9 is equal to.

Question 33
(JEE Mains. 2024)

Remainder when 428^{2024} is divided by 21 is equal to.

Question 34
(JEE Mains. 2022)

Remainder when 2021^{2023} is divided by 7 is equal to.

Question 35

Find the last digit of 3^{104}

Question 36

If $(5+2\sqrt{6})^n = I + f$, where $I \in N, n \in N$ and $0 \leq f < 1$, then prove that $I = \frac{1}{1-f} - f$.

Question 37

(JEE Mains. 2023)

Let $x = (8\sqrt{3}+13)^{13}$ and $y = (7\sqrt{2}+9)^3$. If [t] denotes the greatest integer $\leq t$, then:

1. $[X] + [y]$ is even
2. $[x]$ is odd but $[y]$ is even
3. $[x]$ is even but $[y]$ is odd
4. $[x]$ and $[y]$ are both odd

Question 38

(JEE Mains. 2019)

The coefficient of t^4 in the expansion of $\left(\frac{1-t^6}{1-t}\right)^3$ is?